

NCR MANAGER · QUALITY MANAGEMENT SYSTEM

Validation Support Package

Computer-system-validation template and evidence record
supporting risk-based CSV per GAMP 5.

DOCUMENT

VALIDATION PACKAGE

VERSION

1.0

ISSUED

June 27, 2026

PLATFORM

Kinetic

CONFIDENTIAL · CONTROLLED DOCUMENT

This package supports **computer system validation (CSV)** of MedQMS for a customer's intended use. It is a *template and evidence record*, not a certification. The customer's quality function owns final validation, risk acceptance and release. Approach is risk-based per GAMP 5 / IEC 62304-style rigor as appropriate to the customer's classification of the system.

1. Intended use

MedQMS is intended to manage quality and regulated workflows across the medical-device manufacturing lifecycle — quality-event intake, nonconformance/MRB, CAPA, complaints & vigilance, audits, document & change control, training, supplier quality, ISO 14971 risk management, equipment calibration and management review — with an append-only audit trail and 21 CFR Part 11 electronic signatures. It is a **record-keeping and workflow system**; it does not control a device and is not part of a finished device. It supports, but does not by itself establish, compliance with ISO 13485, 21 CFR Part 820/QMSR, ISO 14971, Part 11 and EU MDR.

2. System requirements (excerpt — URS/FRS)

ID	REQUIREMENT	RISK	VERIFIED BY
URS-01	Capture quality events of all types incl. confidential/anonymous	H	TC-01
URS-02	NC with containment, disposition and MRB approval	H	TC-02
URS-03	CAPA lifecycle blocks closure until actions verified, effectiveness Passed, signatures present	H	TC-03
URS-04	Complaint reportability decision + regulatory deadline tracking	H	TC-04
URS-05	Document/change control auto-assigns training when required	M	TC-05
URS-06	Supplier SCAR cycle with score impact and escalation	M	TC-06
URS-07	ISO 14971 risk file: initial→residual, benefit-risk, post-prod link	H	TC-07
URS-08	Append-only audit trail with actor/time/old→new/reason	H	TC-08
URS-09	Part 11 e-signature with re-authentication, meaning, version, hash	H	TC-09
URS-10	Least-privilege, site/supplier isolation, separation of duties (KSL)	H	TC-10
URS-11	Record-360 traceability across all linked records	M	TC-11
URS-12	Versioned Web API: filter, paginate, idempotent intake, stable errors	M	TC-12
URS-13	Dashboards reflect underlying records (no fabricated aggregates)	M	TC-13
URS-14	No client query exceeds 25 records/page (performance)	M	TC-14

3. Risk assessment (system level, ISO 14971-aligned)

HAZARDOUS SITUATION	HARM	SEV	PROB	CONTROL	RESIDUAL
Regulated record silently overwritten	Loss of traceability / data integrity finding	Serious	Remote	Reason-for-change + append-only audit trail; no update without trail	Low
CAPA closed without effective verification	Recurrence / ineffective correction	Serious	Remote	Closure guard + independent effectiveness signature (SoD)	Low
Reportable complaint misses deadline	Regulatory non-compliance	Serious	Occasional	Reportability field + reg-due clock + overdue escalation/dashboard	Low
Unauthorized access to confidential investigation	Privacy / integrity breach	Serious	Remote	KSL confidential policy; anonymous reporter never disclosed	Low
Signature repudiation	Invalid e-record	Serious	Remote	Re-authentication + bound hash + manifest	Low
Dashboard understates open load (pagination bug)	Missed work	Minor	Remote	De-dup keyset collect; verified counts vs source	Low

4. Test protocol & evidence (IQ / OQ / PQ)

IQ (installation): kapp created; 19 forms created with correct fields; all custom indexes built (background jobs polled to completion); seed loaded. *Evidence:* `install.mjs` console log — 19/19 forms, all index builds "done", ~876/889 records seeded (13 dropped on a benign `handle` uniqueness collision; all showcase records present).

OQ (operational) — executed against `https://ai-labs.kinopsdev.io` :

TC	TEST	RESULT
TC-08	Create record → audit-trail "Created"; edit regulated field → field-diff entry with reason	Pass (round-trip verified)
TC-09	E-signature requires re-auth; manifest entry written with meaning/hash	Pass
TC-11	Record-360 on CAPA-0001 returns full chain (23 nodes/22 edges); supplier reverse-links (SCAR/audit/findings)	Pass
TC-12	<code>/v1/*</code> filter (<code>Risk_Class=Critical</code> → only Critical), pagination (<code>nextPageToken</code>), unknown resource → stable error	Pass
TC-13	Dashboard counts equal source (events 63, capas 26, complaints 40, training 85, ...)	Pass
TC-14	List queries fetch <code>limit=25</code> with keyset paging; no client collect-all	Pass (code + runtime)
TC-03	CAPA closure path & effectiveness states present incl. Failed→Reopened	Pass (data + state model)

PQ (performance qualification — plan): see `PERFORMANCE-TEST-PLAN.md`. Target: 10k active / 100k supported users, millions of records, bounded dashboard latency, indexed selective queries.

5. Deviations

#	DESCRIPTION	DISPOSITION
DEV-01	~13/889 seed inserts failed on a <code>handle</code> unique-index collision	Accepted — sample-data only; no showcase/linked record affected; not a product defect
DEV-02	Browser-driven UI smoke test unavailable in this environment (extension not connected)	Mitigated — all UI data contracts verified via direct API calls + client JS syntax check; manual UI walkthrough is a PQ entry criterion
DEV-03	Workflow task-trees, KSL policies and webhooks are specified as design, not yet instantiated as engine objects	Open — implement per <code>WORKFLOWS.md</code> / <code>ROLES-AND-SECURITY.md</code> during customer configuration; gating logic is enforced in the app pending tree deployment

6. Traceability

Requirements → tests → evidence are linked in §2/§4. Standard-clause coverage is in `REGULATORY-TRACEABILITY.md`. Each executed transition in the running system is itself evidenced in the append-only `audit-trail`.

7. Release approval

ROLE	DECISION	NAME	SIGNATURE	DATE
Validation Lead	Recommend release for customer configuration & PQ	_____	_____	_____
Quality Manager	Approve	_____	_____	_____
Regulatory Affairs	Approve mappings (subject to legal review)	_____	_____	_____

Statement of limitation. This package demonstrates that MedQMS *functions as specified* and *supports* the cited regulatory expectations. It does not assert that any organization is compliant or certified. Compliance depends on the customer's configuration, procedures, data, validation and independent quality/legal review.